Olist E-Commerce Operations Dashboard

Final Project Report

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Assessment	Power BI Practical Examination - Phase 5, Level 2
Project	Multi-Table Data Model Dashboard
Dataset	Brazilian E-Commerce Public Dataset by Olist

Operations-focused sales, delivery and customer-experience analysis using a genuine relational public dataset.

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1. Executive Summary

This project develops an interactive Power BI report for an E-Commerce Operations Manager using the Brazilian E-Commerce Public Dataset by Olist. The solution focuses on three connected decisions: where sales value is generated, where delivery reliability is weakest, and how delivery performance is associated with customer review scores. The project is built as a genuine multi-table model rather than a flat-file dashboard.

The semantic model uses an item-level fact table and four conformed dimensions: customer, product, seller and date. Source files are cleaned and reshaped in Power Query, including two purposeful merge operations. The model then uses explicit DAX measures for revenue, order counts, average order value, delivery reliability, review score and prior-period comparisons. The main report page is deliberately constrained to one screen and is supported by a category drill-through page.

Primary business question

Which product categories and customer regions are driving sales value, and where are late deliveries associated with weaker review scores over time, so an E-Commerce Operations Manager can prioritise logistics and service improvements?

2. Dataset Selection and Qualification

Dataset: Brazilian E-Commerce Public Dataset by Olist. Primary publication: Kaggle - Brazilian E-Commerce Public Dataset by Olist. The data is real commercial e-commerce data that has been anonymised and covers approximately 100,000 orders made in Brazil between 2016 and 2018. The project uses seven related CSV files from the published relational dataset.

Source file	Role in project	Published rows / purpose
olist_order_items_dataset.csv	Main transactional source	112,650 item rows; product, seller, price, freight
olist_orders_dataset.csv	Order header / delivery	99,441 orders; customer, status and timestamps
olist_customers_dataset.csv	Customer dimension	99,441 customer-order records; city/state
olist_products_dataset.csv	Product dimension	32,951 products; category and attributes
olist_sellers_dataset.csv	Seller dimension	3,095 sellers; city/state
olist_order_reviews_dataset.csv	Customer experience	Order-linked review scores and comments
product_category_name_translation.csv	Reference lookup	71 Portuguese-to-English category mappings

The dataset satisfies the multi-table assessment condition because the files share working relational keys: order_id joins items to orders and reviews; customer_id links orders to customers; product_id links items to products; seller_id links items to sellers; and product_category_name links products to the translation lookup. The main fact source alone contains more than 1,000 rows and the order timestamps span well over 12 months.

Source URLs: https://www.kaggle.com/datasets/olistbr/brazilian-ecommerce and https://github.com/olist/work-at-olist-data/tree/master/datasets

3. Stakeholders and End Consumer

Stakeholder	Why the dashboard is relevant
E-Commerce Operations Manager	Primary audience. Needs a concise view of commercial performance, delivery risk and customer experience to prioritise operational action.
Logistics / Fulfilment Lead	Uses state and category delivery indicators to investigate late-delivery concentrations.
Category / Commercial Manager	Uses category revenue, order volume and review performance to identify high-value categories that also need service improvement.
Seller Operations Team	Uses category drill-through and seller-level detail to identify seller groups contributing to performance issues.
Customer Experience Team	Uses review-score patterns to understand where operational reliability is associated with dissatisfaction.

Primary end consumer: E-Commerce Operations Manager. This role was selected because the dashboard combines sales, delivery and review information into an operational decision surface. A more generic audience such as “senior management” would not provide a sufficiently specific basis for choosing what belongs on the canvas.

4. Business Question and Analytical Scope

The report is designed around one decision-focused question rather than a general overview. It asks where value is being generated, where delivery promises are being missed, and whether those missed promises are associated with weaker review outcomes. The question therefore requires product, geography, time, delivery and review fields to work together through the model.

- Commercial performance: Product Revenue, Total Sales Value, Orders, Items Sold and Average Order Value.
- Delivery performance: Delivered Orders, Late Orders, Late Delivery %, On-Time Delivery % and Average Delivery Days.

- Customer experience: Average Review Score, evaluated at order level to avoid overweighting multi-item orders.
- Time comparison: Sales LY, Sales YoY % and Sales YTD using a dedicated date table.
- Decision segmentation: Year, customer state, English product category and delivery status slicers.

5. Power Query and Data Quality

All cleansing and reshaping is performed in Power Query so the process is repeatable and auditable. The original CSVs are not manually edited in Excel. The queries explicitly promote headers, set column types, remove duplicate keys where appropriate, select required fields and merge relational sources.

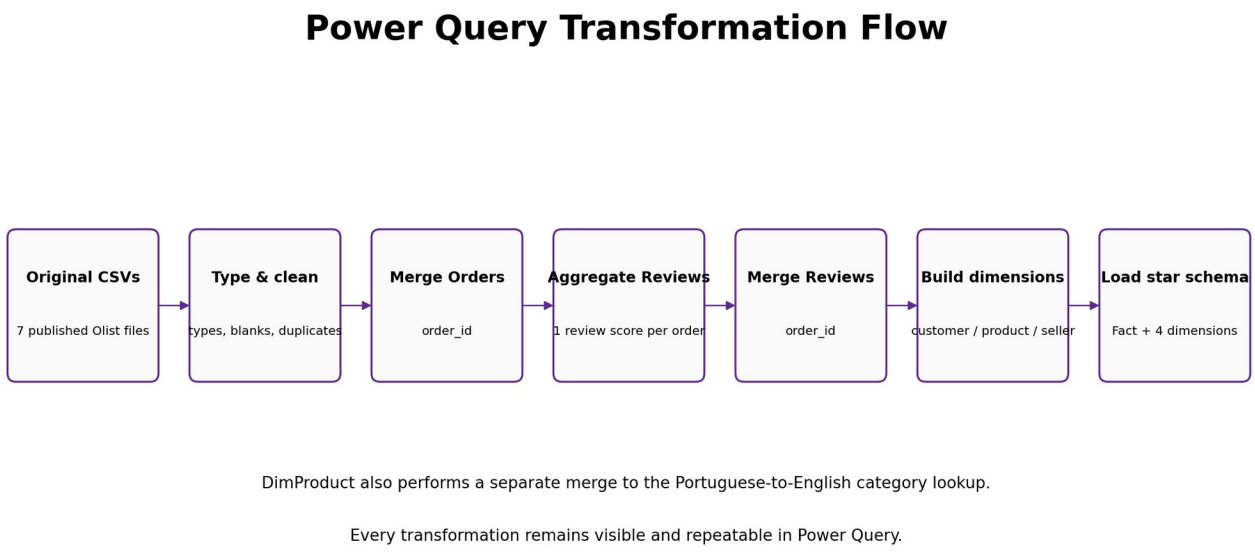


Figure 1. Power Query transformation flow used in the semantic model.

5.1 Fact table construction

1. Load olist_order_items_dataset.csv and assign text, integer, datetime and numeric data types.
2. Remove any duplicate order_id + order_item_id combinations so the fact grain stays one row per order item.
3. Load olist_orders_dataset.csv, retain customer, status and delivery timestamps, then left-merge to order items using order_id.
4. Load olist_order_reviews_dataset.csv and aggregate review_score to one order-level value before merging, avoiding row multiplication where an order has more than one review record.
5. Left-merge the review aggregation to the fact table on order_id.
6. Derive order_purchase_date from the purchase timestamp and create a stable order_item_key.
7. Load the final selected columns into FactOrderItems.

5.2 Dimension construction

- DimCustomer: customer_id, customer_unique_id, customer_city and customer_state; deduplicated on customer_id.
- DimProduct: product fields merged to product_category_name_translation so the dashboard can display English category labels. Missing translations are assigned “Untranslated / Missing”.
- DimSeller: seller_id, seller_city and seller_state; deduplicated on seller_id.
- DimDate: continuous date table from 2016-01-01 to 2018-12-31 with Year, Month, MonthNumber, YearMonth and Quarter.

6. Data Model Design

The model follows a star-schema pattern. FactOrderItems holds the transactional grain and additive numeric values, while descriptive attributes are moved into dimensions. This reduces repeated descriptive data, creates clear one-to-many relationships and keeps filter propagation predictable.

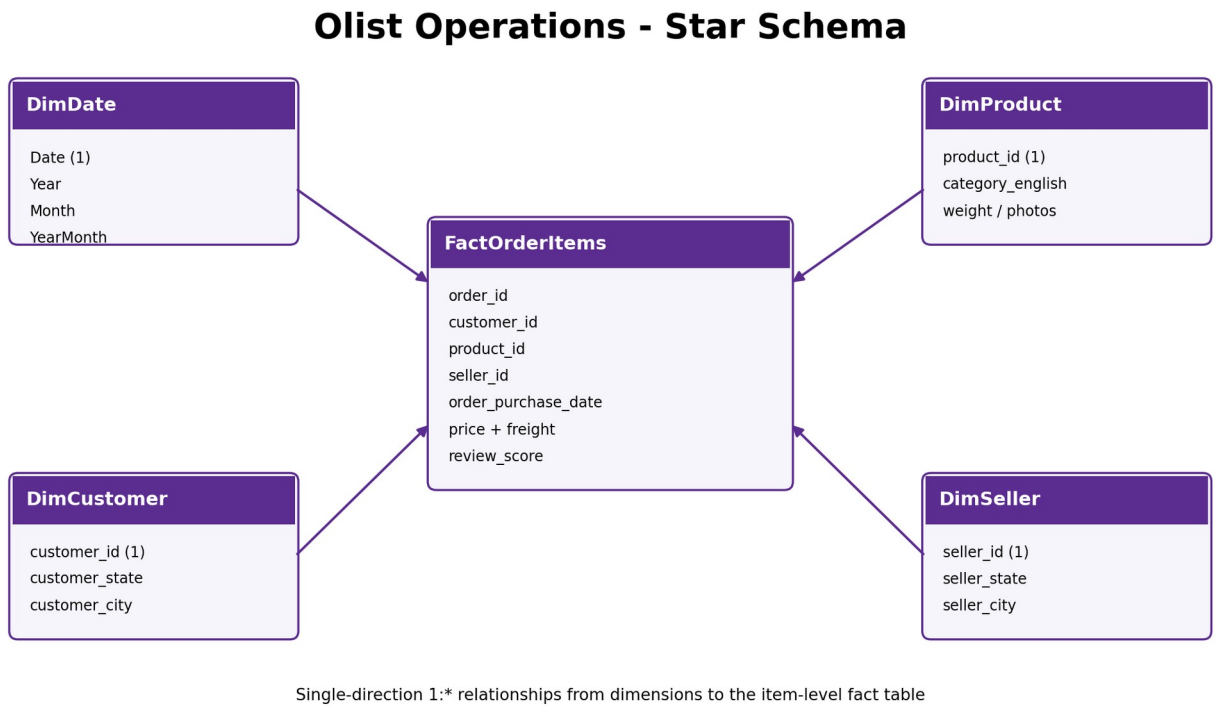


Figure 2. Star-schema relationship design for the Power BI semantic model.

From (many side)	To (one side)	Cardinality	Filter direction
FactOrderItems[customer_id]	DimCustomer[customer_id]	*:1	Single - dimension to fact
FactOrderItems[product_id]	DimProduct[product_id]	*:1	Single - dimension to fact
FactOrderItems[seller_id]	DimSeller[seller_id]	*:1	Single - dimension to fact
FactOrderItems[order_purchase_date]	DimDate[Date]	*:1	Single - dimension to fact

The date table is separate from the fact table because time-intelligence functions require a continuous date axis and should not rely on only the transaction dates that happened to occur in the fact. Month labels are sorted using MonthNumber, and YearMonth is used for trend visuals.

7. DAX Measures and Calculated Column

The solution uses explicit measures rather than relying on implicit aggregations. Measures are preferred where the result should respond to filter context; the calculated Delivery Status column is retained because delivery classification belongs to each transaction row and is useful as a slicer/category.

Measure	Core DAX pattern	Purpose
Product Revenue	SUM(price)	Product value before freight.
Freight Value	SUM(freight_value)	Total freight value.
Total Sales Value	Product Revenue + Freight Value	Headline commercial value.
Orders	DISTINCTCOUNT(order_id)	Unique order count.
Items Sold	COUNTROWS(FactOrderItems)	Item-level transaction count.
Customers	DISTINCTCOUNT(DimCustomer[customer_unique_id])	Unique marketplace customer count using customer_unique_id.
Average Order Value	DIVIDE(Total Sales Value, Orders)	Average sales value per order.
Delivered Orders	CALCULATE(Orders, status = delivered)	Delivered-order denominator.
Late Orders	CALCULATE(Orders, Delivery Status = Late, delivered)	Explicit CALCULATE filter measure.
Late Delivery %	DIVIDE(Late Orders, Delivered Orders)	Delivery reliability KPI.
Average Review Score	AVERAGEX(VALUES(order_id), order score)	Order-level review score avoiding item weighting.
Average Delivery Days	AVERAGEX by order + DATEDIFF	Average purchase-to-delivery elapsed days.
Sales LY	CALCULATE + SAMEPERIODLASTYEAR	Prior-year sales value.
Sales YoY %	DIVIDE(current - LY, LY)	Year-on-year change.
Sales YTD	TOTALYTD	Year-to-date time intelligence.
On-Time Delivery %	1 - [Late Delivery %]	Complementary delivery-reliability KPI.
Dashboard Title	Text measure	Dynamic presentation title used by the report canvas.

Measure	Core DAX pattern	Purpose
Dashboard Subtitle	Text measure	Dynamic presentation subtitle describing audience, source and period.

7.1 Calculated column

Delivery Status classifies each row as “Not Delivered”, “Late” or “On Time” by comparing actual delivery date with the estimated delivery date. This is a justified calculated column because the classification is a row-level attribute that is reused for slicing, chart categories and filtered measures.

8. Dashboard Design and Interactivity

The main page is an executive operations page designed to fit a single screen. It places the decision controls first, headline KPIs next, then trends and diagnostic visuals. The second page is a detail page intended for drill-through from a selected product category.

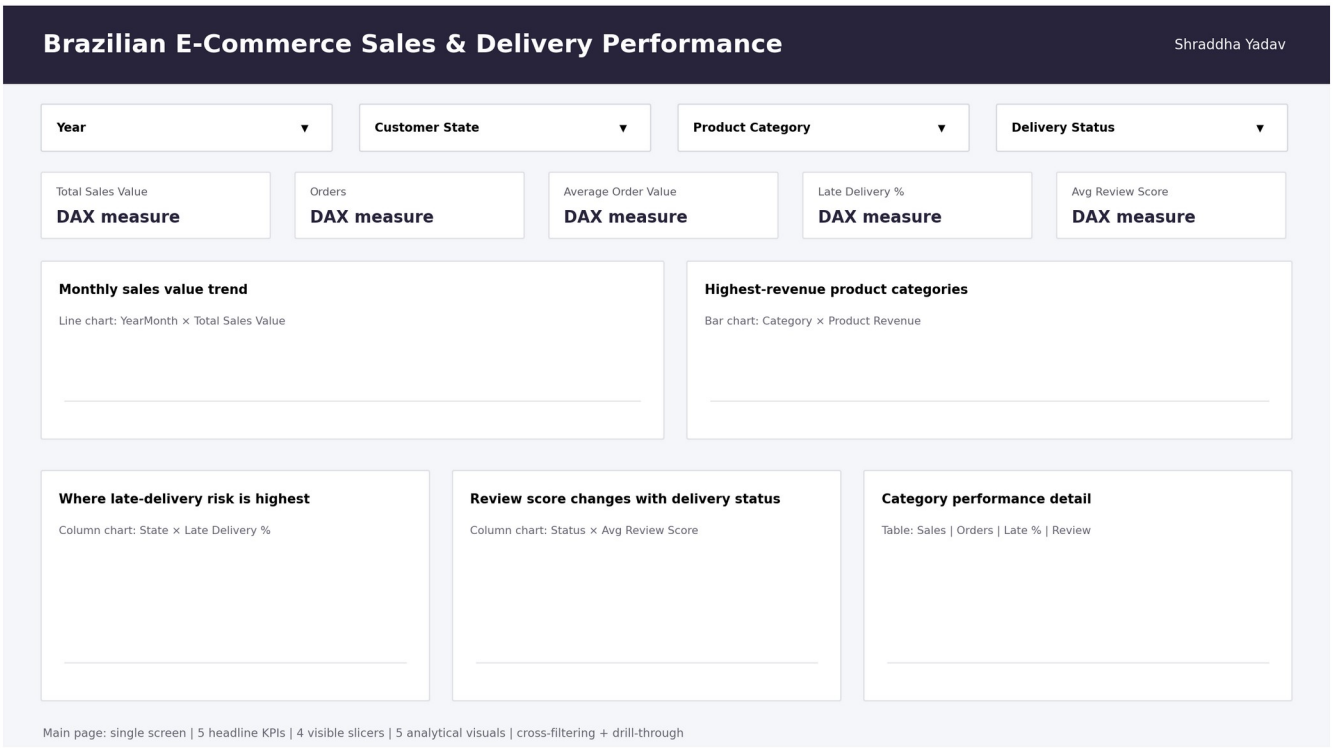


Figure 3. Final dashboard design and visual placement. Live values are provided by the DAX measures when the project is refreshed in Power BI Desktop.

Visual / control	Fields / measure	Decision supported
Year slicer	DimDate[Year]	Restrict analysis to a year.
Customer State slicer	DimCustomer[customer_state]	Compare destination markets.

Visual / control	Fields / measure	Decision supported
Product Category slicer	DimProduct[product_category_name_english]	Focus operational review on a category.
Delivery Status slicer	FactOrderItems[Delivery Status]	Separate on-time, late and not-delivered activity.
5 KPI cards	Sales, Orders, AOV, Late %, Review	Immediate commercial and service health.
Line chart	YearMonth x Total Sales Value	Detect growth and seasonal movement.
Bar chart	Category x Product Revenue	Find the categories driving value.
Column chart	State x Late Delivery %	Locate geographic delivery risk.
Column chart	Delivery Status x Average Review Score	Compare satisfaction by service outcome.
Table	Category + Sales + Orders + Late % + Review	Read multi-metric category detail.
Category drill-through	Selected category context	Investigate revenue by state and seller performance.

Cross-filtering is left enabled so selecting a state, category or delivery status changes the rest of the page. The drill-through page accepts Product Category context and provides category-specific KPIs, monthly revenue, state revenue and seller-level detail.

9. Evidence of the Eight Required Techniques

Required technique	Where it is demonstrated
Power Query cleaning	Header promotion, explicit types, duplicate handling, column selection and null-aware transformations in the semantic-model partitions.
Merge or append	FactOrderItems merges Orders and aggregated Reviews on order_id; DimProduct merges the category translation table.
Relationships	Four working *:1 relationships connect the fact table to Customer, Product, Seller and Date dimensions.
Proper date table	DimDate is continuous for 2016-2018, categorized as Time and related to order_purchase_date.
Calculated column	FactOrderItems[Delivery Status] classifies actual versus estimated delivery.
DAX measures	Sixteen analytical measures plus two presentation text measures, including CALCULATE filter measures and ratios.
Time intelligence	Sales LY, Sales YoY % and Sales YTD use DimDate.

Required technique	Where it is demonstrated
Interactivity	Four visible slicers, cross-filtering and a Product Category drill-through page.

10. Validation and Quality Checks

- Grain check: FactOrderItems is one row per order item after duplicate removal.
- Join check: order_id, customer_id, product_id and seller_id are consistently typed as text on both sides of each relationship.
- Review join check: reviews are reduced to one order-level score before joining, preventing accidental multiplication of item rows.
- Dimension check: dimension keys are deduplicated before model load so the one side of each relationship is unique.
- Date check: DimDate is continuous and covers the entire analytical period.
- Measure check: order-level measures use DISTINCTCOUNT or VALUES(order_id) where item-level duplication would otherwise bias results.
- Filter check: relationships are one-directional, reducing ambiguity and making slicer effects predictable.
- Presentation check: currency, percentage and decimal formats are assigned at measure level.

11. Key Insights and Business Interpretation

The model is designed to quantify the following patterns in the Olist data. These interpretations are consistent with independent public analyses of the same published dataset and provide useful validation targets when the Power BI file is refreshed.

Insight	Business interpretation
Revenue is concentrated in a small group of categories.	Public Olist analyses consistently identify Health & Beauty, Watches & Gifts, Bed/Bath/Table and Sports & Leisure among the strongest revenue categories. The dashboard therefore combines category value with delivery and review metrics rather than treating revenue alone as success.
Late delivery is strongly associated with lower review scores.	Independent analyses report average review scores around 4.2-4.3 for on-time orders versus about 2.6 for late orders. The operational priority is therefore not simply fast delivery but meeting the promised date.
Delivery risk varies materially by destination state.	Published analyses show substantial differences in late-delivery rates between Brazilian states, so state is exposed as both a slicer and diagnostic visual.
The sales history shows rapid platform growth through 2017 and into 2018.	The monthly trend and prior-year measures help distinguish growth from one-off monthly movement and provide context for operational workload.

Insight	Business interpretation
A high-revenue category can still require operational attention.	The category drill-through deliberately combines revenue, orders, late-delivery percentage and review score so the user can distinguish commercially valuable categories from operationally healthy ones.

Interpretation rule
The dashboard shows association, not causation. A lower review score on late orders is operationally important, but the model does not prove that lateness is the sole cause of a customer's rating.

12. Limitations and Responsible Interpretation

- The dataset covers Olist marketplace activity from 2016-2018 and should not be presented as current Brazilian e-commerce performance.
- The customer_id field is order-linked; customer_unique_id is the better identifier for repeat-customer analysis, which is outside this dashboard's primary operational scope.
- Review text is not used in this report; only review_score is included in the analytical model.
- Orders without a recorded delivered date are classified as Not Delivered and are excluded from the delivered-order denominator for Late Delivery %.
- The model uses published commercial data that has been anonymised, so no individual customer profiling is attempted.

13. Conclusion

The finished solution meets the purpose of the Level 2 assessment by treating the semantic model as the foundation of the analysis. It demonstrates multi-table Power Query shaping, a star schema with controlled filter direction, a dedicated date dimension, explicit DAX measures, row-level calculated logic, time intelligence, interactive filtering and drill-through. The resulting report is intentionally focused on the decisions of an E-Commerce Operations Manager rather than on maximising the number of visuals.

14. References

- Olist. Brazilian E-Commerce Public Dataset by Olist. Kaggle: https://www.kaggle.com/datasets/olistbr/brazilian-ecommerce
- Olist public dataset mirror: https://github.com/olist/work-at-olist-data/tree/master/datasets
- Microsoft Learn. Power BI Desktop projects / PBIR and semantic-model project formats: https://learn.microsoft.com/power-bi/developer/projects/projects-overview
- Independent validation examples used only to cross-check broad patterns: Data Science Projects - Olist Delivery & Satisfaction; public Olist SQL analyses available on GitHub.